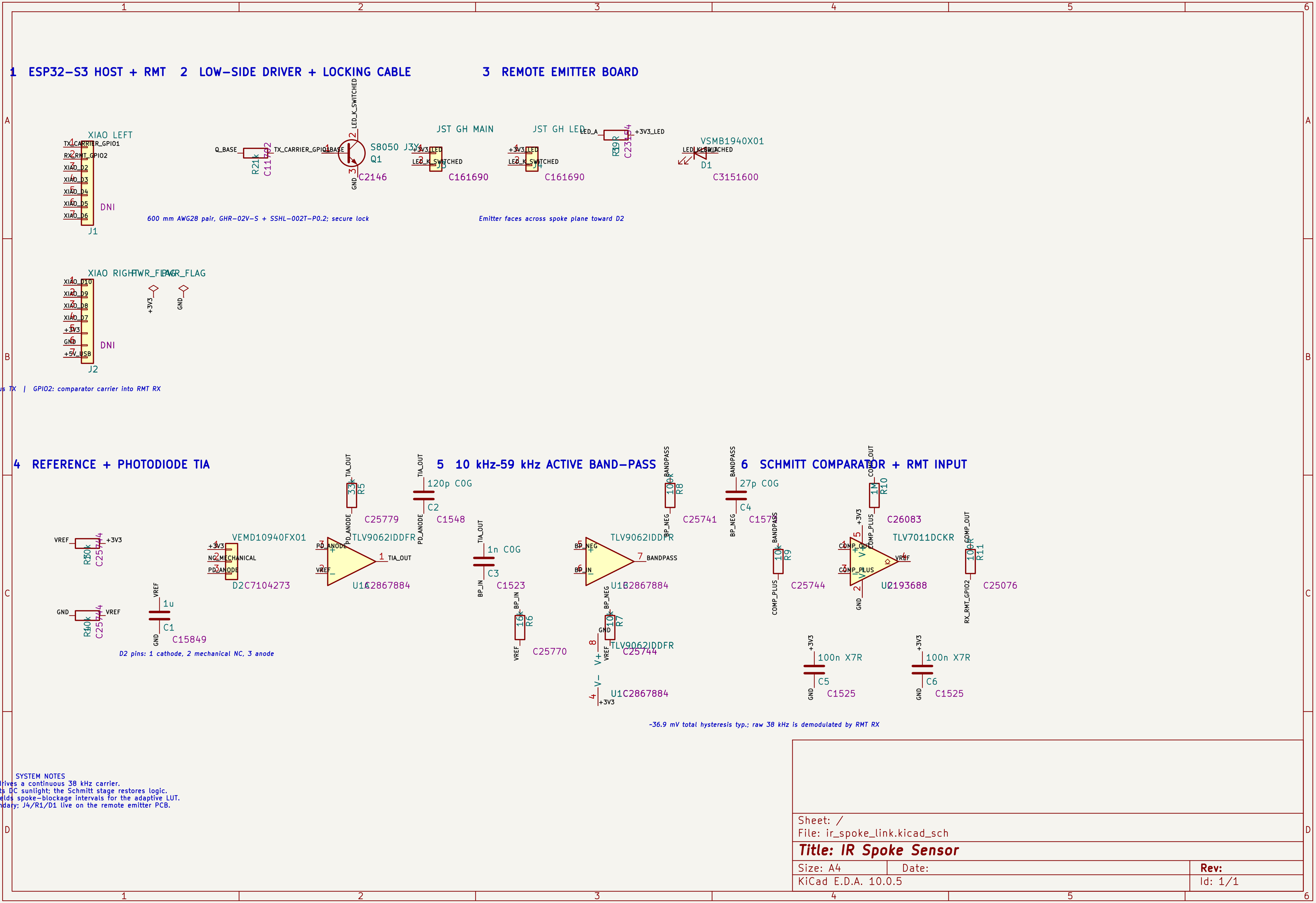
1 ESP32-S3 HOST + RMT
2 LOW-SIDE DRIVER + LOCKING CABLE
3 REMOTE EMITTER BOARD
XIAO LEFT
TX_CARRIER_GPIO1
RX_RMT_GPIO2
XIAO_D2
XIAO_D3
XIAO_D4
XIAO_D5
XIAO_D6
DNI
J1
Q_BASE
R2
1k
C11702
TX_CARRIER_GPIO1
BASE
LED_K_SWITCHED
S8050 J3Y
Q1
C2146
GND
JST GH MAIN
+3V3_LED
LED_K_SWITCHED
J3
C161690
JST GH LED
+3V3_LED
LED_K_SWITCHED
J4
C161690
LED_A
R1
39R
C23144
+3V3_LED
VSMB1940X01
LED_A
LED_K_SWITCHED
D1
C3151600
600 mm AWG28 pair, GHR-02V-S + SSHL-002T-P0.2; secure lock
Emitter faces across spoke plane toward D2
XIAO RIGHT
PWR_FLAG
PWR_FLAG
+3V3
GND
XIAO_D10
XIAO_D9
XIAO_D8
XIAO_D7
+3V3
GND
+5V_USB
DNI
J2
kHz continuous TX | GPIO2: comparator carrier into RMT RX
4 REFERENCE + PHOTODIODE TIA
5 10 kHz-59 kHz ACTIVE BAND-PASS
6 SCHMITT COMPARATOR + RMT INPUT
VREF
R3
10k
C25744
+3V3
GND
R4
10k
C25744
VREF
VREF
1u
C1
C15849
GND
VEMD10940FX01
+3V3
NC_MECHANICAL
PD_ANODE
D2
C7104273
D2 pins: 1 cathode, 2 mechanical NC, 3 anode
TIA_OUT
R5
33k
C25779
PD_ANODE
TLV9062IDDFR
PD_ANODE
VREF
TIA_OUT
U1A
C2867884
TIA_OUT
120p C0G
C2
C1548
PD_ANODE
TIA_OUT
1n C0G
C3
C1523
BP_IN
BP_IN
R6
16k
C25770
VREF
TLV9062IDDFR
BP_NEG
BP_IN
BANDPASS
U1B
C2867884
BP_NEG
R7
10k
GND
VREF
C25744
TLV9062IDDFR
V+
V-
U1C
C2867884
+3V3
BANDPASS
R8
100k
C25741
BP_NEG
BANDPASS
27p C0G
C4
C1575
BP_NEG
BANDPASS
R9
10k
C25744
COMP_PLUS
COMP_OUT
R10
1M
C26083
+3V3
COMP_PLUS
TLV7011DCKR
COMP_OUT
COMP_PLUS
VREF
U2
C2193688
GND
COMP_OUT
R11
100R
C25076
RX_RMT_GPIO2
+3V3
100n X7R
C5
C1525
GND
+3V3
100n X7R
C6
C1525
GND
~36.9 mV total hysteresis typ.; raw 38 kHz is demodulated by RMT RX
SYSTEM NOTES
• RMT TX drives a continuous 38 kHz carrier.
ete AFE rejects DC sunlight; the Schmitt stage restores logic.
er removal yields spoke-blockage intervals for the adaptive LUT.
a cable boundary; J4/R1/D1 live on the remote emitter PCB.
Sheet: /
File: ir_spoke_link.kicad_sch
Title: IR Spoke Sensor
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/1